HOTLINE FIX

Group: 15 Intake and Exhaust	**Original Date:** 16 July 2010	**No.** ML15-005
Model: ML-MN Triton – 4D56 Only PB Challenger.	**Subject:** Check engine lamp on **with** clogged boost sensor nipple (4D56 eng)	
Revised on 26 August 2011 **Revised on** ..... (date) **Revised on** ..... (date)	D R Budden Manager National Service	
MMAL Use Only P3883 TSL 201003161700	**Document consists of 3 page**	

(1) **Description**: The check engine lamp turns on (with DTC P0106 or other related DTC) **and** the boost sensor nipple on the inlet manifold is found to be clogged with soot.
The purpose of this revision is to add a diagnosis procedure to determine whether the MAP (Manifold Absolute Pressure) sensor is faulty as well as having a blocked MAP sensor nipple.

Note: The DTC **MUST** be recorded on any warranty repair order.

(2) **Possible Cause**: On early production vehicles the boost sensor nipple is located downstream of the EGR valve. This can result in soot deposits clogging the nipple and logging a boost pressure related DTC.

(3) **Improvement**: As a running change the boost sensor port has been re-located upstream of the EGR valve.

Note: This is for information only, vehicle upgrades **are not approved** under standard warranty.

Model Code	Name	Engine Type	Effective Build Date	Engine Number
KA4T	Triton (2WD)	4D56 NP	20/01/2010	4D56UCBW8649
KB4T	Triton (4WD)	4D56 HP	03/02/2010	4D56UCBX4698
KH4W	Challenger	4D56 HP	04/02/2010	4D56UCBX4667

(4) **Field Fix/Action**: On a customer complaint basis only, with check engine lamp on take the following action.

a) Check the DTC and record on the repair order.
b) Conduct the "**Diagnosis procedure to determine whether the MAP sensor nipple is clogging and/or the MAP sensor is faulty"** detailed in paragraph 7.

(5) **Parts Information**
Please refer to the table below for the necessary parts.

Part Name	Part No. (New)	Part No. (Old)	Remarks
Inlet manifold	1540A417	1540A344	
Hose assy. Vacuum (Normal Power)	2036A320	2036A227	Models(Normal Power)
Hose assy. Vacuum (Hi-Power)	2036A319	2036A226	Models(High Power)
Engine cover stay *(some models only)*	1003A193	1003A015	Vehicles produced before July 2009 only
Gasket, Throttle body	1450A147		Not re-usable
Gasket, EGR valve	1582A039		Not re-usable
Gasket, EGR pipe（EGR cooler side）	MR161679		Not re-usable
Gas et, Inlet manifold	1540A0 9		Not re-usable
Gasket, Inlet manifold	1540A050		Not re-usable
Gasket, Air temperature sensor	MF660064		Not re-usable
MAP Sensor	1865A035		

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Note: This document should be forwarded direct to the Service Manager for action and distribution to all affected staff

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(6) **Warranty Information**
Standard terms and conditions apply

Quality Information				
Causal Part	Position Code	A-Code	B-Code	C-Code
1540A417	151110	57 – Clogged	04 (Poor output)	As per customer complaint

Rectification Procedure

Repair Information			
Position Code	Work Code	Description	Time
151110	10	Inlet manifold replacement	Refer WIS

Warranty data below for MAP sensor replacement (if required)

Repair Information			
Position Code	Work Code	Description	Time
138322	10	Manifold Absolute Pressure Sensor replacement	Refer WIS

Remarks	(list the actual 5 character DTC shown on the freeze frame data, in the WIB 2009-01 format)
Record DTC ___	

(7) **Diagnosis procedure to determine whether the boost nipple is clogging and/or the MAP sensor is faulty.**

Step 1: Check for DTC's in the Engine ECU using MUT III
If DTC's P0107 (MAP sensor circuit low input) or P0108 (MAP sensor circuit high input) are stored replace the intake manifold and the MAP sensor as the MAP sensor is faulty.
If DTC's P0106 (MAP sensor range/performance problem), are stored, go to Step 2 below.

Step 2: Inspect MAP sensor using MUT III data list.
With the engine off, view the Engine ECU data list using MUT III and compare the Barometric Pressure Sensor and the Manifold Absolute Pressure Sensor. If the MAP sensor is operating correctly the two signals should read the same.
If MAP sensor value varies from the Barometric pressure sensor value, replace the intake manifold and the MAP sensor as the MAP sensor is faulty.
If MAP sensor value is the same as the Barometric pressure sensor value go to Step 3.

Step 3: Flick the MAP sensor using your finger and check whether output voltage varies.
If the output voltage **does not** vary the MAP sensor is in good condition therefore only replace the intake manifold.
If the output voltage **does** vary, the MAP sensor is faulty so replace the intake manifold and the MAP sensor.

Note: In the event of this condition occurring on a vehicle manufactured after the effective build date, submit a DQR and ensure the relevant freeze frame data is attached.

(8) **Replacement Procedures.**

(1) Replace the inlet manifold, vacuum hose and engine cover stay with modified ones.
(2) Connect the vacuum hose to the MAP sensor and nipple of the inlet manifold. (refer to the figure below)
(3) Please refer to the Workshop Manual for the inlet manifold replacement procedure.
(4) Non re-usable parts (gaskets) should be replaced with new ones.

